PROJECT LESSON · TECHNOVATION GIRLS 2026

Kesararam With Digital (KWD)
A Bilingual Road-Safety Learning Platform

A free web application that teaches Cambodian road rules and traffic-sign theory in Khmer and English — through structured lessons, flashcards, quizzes, live classroom games, and an AI study assistant.

Team #50777 | Siem Reap, Cambodia | kesararamwithdigital.tech

AGENDA

What we will cover today

- **The problem** — why road-safety education is hard in Cambodia
- **Our solution** — what KWD is and who it serves
- **Key features** — learn, play, compete, ask AI
- **How a student uses it** — step by step
- **How a teacher uses it** — live classroom games
- **The technology** — how the app is built
- **System architecture** — how the pieces connect
- **Impact & what is next**

THE PROBLEM

Why this matters

Road risk is high

Cambodia has one of the region's highest road-crash rates. Many crashes trace back to weak knowledge of signs and rules.

Study material is scarce

Official theory content is limited, often paper-only, and rarely available in clear, modern Khmer-first digital form.

Learning is not engaging

Rote memorization is boring. Students need interactive practice, feedback, and motivation to keep going.

Our question: How can we make correct road-safety knowledge free, fun, and accessible to every Cambodian student — in their own language?

OUR SOLUTION

What is KWD?

KWD is a free, bilingual (Khmer + English) web platform that turns road-safety theory into an interactive learning experience.

Learn

Structured lessons across 8 knowledge areas with signs, images, and references.

Play

Solo quizzes and category practice with instant feedback.

Compete

Live classroom games hosted by teachers, with a leaderboard.

Ask AI

"Suar AI" answers road-rule questions in Khmer or English.

Who it is for

Students, teachers, and new drivers preparing for the license theory test.

Languages

Full Khmer + English everywhere; switch with one tap.

Cost

Free for students. Teachers can upgrade for advanced class tools.

CONTENT

8 knowledge areas

All learning content is organized into eight categories, managed by admins and reviewed before publishing.

Road Signs

Warning, prohibitory, mandatory, and information signs.

Road Traffic

Traffic laws and rules of the road.

Priority Rules

Right of way at junctions and crossings.

Driving

Safe driving behaviour and technique.

Technical

Vehicle basics and maintenance.

Loading People & Goods

Safe carrying of passengers and cargo.

General Knowledge

Broader road-safety awareness.

Emergency

What to do during incidents.

USER JOURNEY · STUDENT

Learn road safety in three simple steps

01

Browse Categories

Pick one of the 8 areas and study lessons, signs, and references.

→

02

Study & Practice

Use flashcards, quizzes, and the Suar AI assistant in Khmer or English.

→

03

Join Live Challenges

Compete in teacher-hosted sessions, earn points, climb the leaderboard.

Motivation loop: answer questions → earn points → rank on the leaderboard → unlock rewards. This keeps students coming back to practice.

HOW IT WORKS · LIVE CLASSROOM GAME

Real-time multiplayer quiz

Teacher side

- Creates a session and picks categories & difficulty
- A unique **game code** + QR code is generated
- Teacher presents questions on a big screen
- Controls the pace; correct answers reveal automatically

Student side

- Scans the QR code or enters the game code
- Joins the lobby on their phone (full-screen)
- Answers each question against a timer
- Sees instant feedback and live ranking

The technology behind it: Pusher (websockets) sends questions and scores to every device at the same moment, so the whole class stays in sync in real time.

FEATURE · SUAR AI

An AI study assistant

What it does

- Answers questions about signs, rules, and driving
- Explains in **Khmer or English**
- Helps students who are stuck while studying

How it works

- A large language model (LLM) processes the question
- The app sends the prompt securely from the server
- The answer is streamed back to the student

TECHNOLOGY

How the app is built

LAYER	TECHNOLOGY	WHY WE USE IT
Framework	Next.js 15 + React 19	Fast, modern web app with pages + server logic in one place
Language	TypeScript	Catches errors early; safer, clearer code
Styling	Tailwind CSS v4	Consistent brutalist design system
Database	Neon (PostgreSQL)	Reliable cloud SQL database
Data access	Drizzle ORM	Type-safe database queries
Auth	JWT (jose)	Secure login & sessions
Real-time	Pusher	Live multiplayer game sync
Media	Cloudinary	Stores & optimizes images/audio
AI	LLM (Groq / OpenAI)	Powers the Suar AI assistant
Maps	Mapbox	Live traffic map on the home page

SYSTEM ARCHITECTURE

How the pieces connect

Browser

Student / teacher device. Renders pages, runs the live game UI.

→

Next.js Server

Server actions + API routes handle all logic and writes.

→

PostgreSQL

Stores users, content, questions, scores — 36 tables.

Pusher

Pushes live questions & scores to every player at once.

Cloudinary

Serves sign images and audio quickly.

AI + Mapbox

External services for the assistant and live traffic map.

Security: All database writes go through the server. Secret keys (database, AI, payments) never reach the browser. Backend dashboards are hidden from the public.

ROLES & ACCESS

Who can do what

Student

Learns, plays solo, joins live games, tracks points & rewards.

Teacher

Manages classes, hosts live sessions, sees class results.

Staff

Creates and submits learning content for review.

Admin

Approves content, manages users, oversees the whole system.

Content workflow: Staff create lessons/questions → Admin reviews → once approved & published, students can study and play with them. This keeps information accurate and trustworthy.

DESIGN IDENTITY

Road safety, made into a UI

Brutalist + traffic-sign style

- Sharp corners, solid black borders
- Hard offset shadows, no blur, no gradients
- Caution-tape accents (yellow + black)
- Library icons only — no emoji

Color used like a road sign

Yellow · CTA / warning

Red · stop / danger

Green · go / correct

Blue · info

Orange · caution

Purple · premium

Every color carries a meaning, just like real road signage.

IMPACT

Why KWD makes a difference

Accessible

Free, mobile-friendly, and Khmer-first so anyone can learn.

Engaging

Games, points, and live competition keep students motivated.

Trustworthy

All content is reviewed by admins before students see it.

Goal

Help more Cambodian students learn correct road rules, pass the theory test with confidence, and ultimately make the roads safer for everyone.

WHAT'S NEXT

Roadmap

Short term

- Grow the question bank across all 8 areas
- More classroom game modes
- Offline-friendly study and richer flashcards

Long term

- Partner with schools and driving centres
- Deeper analytics for teachers
- Reach students across all of Cambodia

THANK YOU

Questions?

Kesararam With Digital — making road-safety education free, bilingual, and fun for every Cambodian student.

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